M1 — Problem and customer validation

1. Moving companies in Luxembourg need to estimate the volume of a customer's belongings before they can prepare an accurate moving quote. This requires an employee to visit the customer's home, inspect the furniture and boxes, and manually estimate the total volume. This process takes staff time, costs money to send people to measure everything each time there is a relocation needed, creates scheduling delays, and adds time for customers who want to organise an international relocation. These difficulties may be particularly relevant in Luxembourg, where there is a high demand for moves into, out of, and within the country. A photo-based AI tool could allow customers to photograph their flats and automatically estimate the volume in cubic metres of all the furniture and prepare a quote for the moving company to review. The key problem to validate is whether this approach can reduce the time and effort required to prepare quotes while producing estimates accurate enough for practical use.

2. There are three distinct groups to consider: customers needing a relocation, people working in a moving company and the businesses paying for it (which are the actual moving companies).

Primary users: Moving company staff

Employees responsible for preparing moving quotes would use the AI-generated inventory and volume estimate to assess a customer's move more quickly. They could review the result and adjust it before sending a final quote.

Paying customers: Moving companies

Small and medium-sized moving companies operating in Luxembourg are the initial target market. They could pay a subscription or a fee per estimate if the tool reduces staff workload and makes it possible to handle more quote requests.

End users: People planning a move

Individuals and families moving within Luxembourg, arriving from abroad, or leaving the country could photograph their belongings using a phone. They would benefit from such process without having to deal with arranging a home visit.

It is also worth mentioning that moving company staff would either:

- have less work to do

- have different work to do (instead of driving to different places they would review the pictures and possibly send feedback if any corrections were needed)

- would lose their jobs

So the money invested in company staff could be put into paying for using such a AI system.

Target segment and market size: The initial target market is small and medium-sized moving companies operating in Luxembourg that regularly prepare household moving quotes. Individuals and families moving into, out of, or within Luxembourg would be the end users supplying photos of their belongings. STATEC recorded 24,985 arrivals and 17,141 international departures in Luxembourg in 2025, indicating a substantial level of cross-border population movement. These numbers are not a direct estimate of the market for this product but more like a demand indicator. We can't accurately

state the number of companies that would be willing to buy such product without other form of information, for example: interviews.

3.Interviews (bullet points, notes and details are below).

1st:

1- AI could reduce manual listing, but customers should be able to add missing items and correct quantities.

2- Room-by-room filming instructions should include cellars and storage areas to reduce omissions**.**

3- Customers should be able to update the number of boxes as packing progresses.

4- Clear filming instructions and information about video access, storage and deletion could support adoption.

5- Uncertain dimensions should be flagged, and the moving company should review the estimate, access conditions and pricing before confirming the quotation.

2nd:

1- Complete inventories, dimensions and access information are essential for preparing reliable quotations.

2- Missing items and unclear access conditions can lead to additional checks and inaccurate estimates.

3- AI could help create an initial inventory and reduce manual information collection.

Staff would still need to verify uncertain measurements, unusual items and the final quotation.

4- Adoption would depend on demonstrated time savings, reliable results and a price justified by those benefits.

4. Three existing alternatives:

• In-person visits by company's estimators:

Fall short because of the costs of transport (company car, petrol) and covering the worker's wage by the company. Moreover it occupies the schedule for other visits.

• Hiring freelance estimators

High costs of visits, relying on independent workers and their limited time

• Web forms with manual customer data entry:

Cheaper than hiring workers for this job, but the company needs to be aware of the clients' errors which can lead to the matching of the wrong truck

• 3D scanning apps:

They require the customers to perform a slow, thorough scanning proces + the apps don't itemise the scanned furniture or calculate total transport volume

5. For moving companies who spend money on their workers wage to investigate customer's apartments, meanwhile losing opportunities for extra income due to full schedule, our AI app is a quoting automation tool that categorises furniture and estimates its volume from user-submitted photos, unlike hiring workers to measure the volume themselves.

6. Why AI

AI can identify furniture and boxes in a customer's video and turn these observations into a structured inventory. Without AI, a moving-company employee would need to review the footage manually or conduct an on-site visit, which takes time. The inventory supports volume estimation using reference dimensions and helps prepare a preliminary quotation. Customers confirm detected items and quantities, while the moving company validates the estimate before issuing the quotation.

Konrad Sarba, Oliwer Miller, KAMKOUM TANGUE Maurice Segain

OMK Labs

Interview notes:

Interview 1 with an anonymous person Luxembourg who recently moved.

This interview focuses on the experience of a person who recently moved. It aims to understand how they organised their move, obtained quotations and communicated information about their belongings. It also explores the difficulties they encountered and their views on an AI-assisted video tool, particularly its usefulness, ease of use and privacy implications.

QUESTIONS:

Q. Could you briefly describe your recent move? Did you use a moving company, move independently, or combine both?

A: I moved between two apartments in Luxembourg. I packed smaller belongings myself and hired a moving company to transport the furniture and boxes.

Q. What information did you have to provide to obtain a quotation, and how did you provide it?

A: I provided both addresses, the moving date, the floors and lift availability, along with a list of furniture and an approximate number of boxes. I sent photos by email, and the company contacted me to clarify some dimensions.

Q. Approximately how much time did you spend listing your belongings, providing information and arranging assessments? Were there any related expenses?

A: I did not measure the time precisely, but I had to prepare the list, take photos and answer follow-up questions on several occasions. There was no separate assessment fee, but it took time to gather the information.

Q. What was the most difficult part of describing your belongings or arranging the move? Could you give a specific example?

A: Estimating the number of boxes was difficult because I had not finished packing. I also did not know which furniture dimensions mattered, so I had to ask the company and take additional measurements.

Q. Were any belongings forgotten, dimensions misunderstood or access conditions overlooked? If so, what happened?

A: I initially forgot some items stored in the cellar. After adding them, the company reviewed the inventory and checked whether the planned transport capacity was sufficient. This showed me how easily a written list can be incomplete.

“ Thank you for sharing your experience. You mentioned difficulties estimating box quantities, identifying relevant measurements and remembering items in storage areas. We would now like to hear your views on a possible approach: customers film each room and storage area, and AI proposes an inventory that they can review and correct. The tool would then estimate an approximate volume and help the moving company prepare a preliminary quotation. Thinking back to your move, where might this have helped, and what concerns would you have? ”

Q. Thinking back to your move, which steps could this tool have made easier, and which would still have required help from the moving company?

A: It could have helped me create the initial furniture inventory instead of writing everything down. However, I would still have needed guidance on measurements and access conditions. It would not have solved the uncertainty about the final number of boxes before packing was finished.

Q. How would you feel about filming each room and storage area? What could make this difficult or uncomfortable?

A: I would consider it if the filming instructions were clear. I would want to avoid recording personal documents, valuables or other people. Before uploading, I would also want to know who could access the video and when it would be deleted.

Q. If the tool generated an inventory, how would you check it? What would help you identify missing or incorrectly counted items?

A: I would review the inventory room by room and compare it with my belongings. A checklist reminding me to include the cellar and other storage areas would help. I would also want to edit quantities, add missing items and see which dimensions needed confirmation.

Q. What would make you comfortable using the estimated volume and preliminary quotation to plan your move?

A: I would want the moving company to review the information and explain what was included in the quotation. Uncertain measurements and possible additional charges should be clearly identified. I would use the preliminary quotation for planning, but I would still want confirmation from the company before booking.

2nd Interview with a domain experts :

Interviewee: **Jordan Samen, Owner of a moving company in Luxembourg**

Date: **11 October 2026**

Duration: **30 min**

Interviewer: **Maurice Kamkoum**

“ Hello, I am a student at the University of Luxembourg. Our team is studying how moving companies prepare quotations and exploring whether video-based artificial intelligence could make this process easier. This interview will take approximately 20–30 minutes. May I take notes and include an anonymised summary in our coursework? You may skip any questions if you wish.”

QUESTIONS:

Q- What is your role in the company, and are you involved in preparing moving quotations?

A : I coordinate moving requests and prepare quotations. I collect information from customers, estimate the resources needed and discuss difficult cases with our operational team

Q- Could you walk me through how you prepared your most recent quotation?

A: The customer contacted us by email. We asked for the moving addresses, preferred date, an inventory and access details. After reviewing the photos, we requested additional measurements, estimated the volume and prepared the quotation.

Q- What information do you need from the customer before you can provide a quotation?

A: We need the inventory, approximate dimensions, addresses, floors, lift availability, parking conditions and any packing or dismantling services .

Q- When do you use an on-site visit, photos, a video call or another method?

A: Photos may be sufficient for a small move. For a larger or more complicated move, we prefer a video call or an on-site visit.

Q- How much staff time does one quotation typically require, including travel and administration?

A: A straightforward quotation can be prepared relatively quickly when the customer provides complete information. Follow-up questions and on-site visits take longer. Travel and administration should be included when measuring the total time.

Q- Approximately how much does preparing one quotation cost your company? An estimate or range is fine.

A: We do not calculate a separate cost for every quotation. The main expenses are staff time and travel. We could estimate the cost using the time spent, our internal hourly staff cost and any travel expenses

Q- What mistakes or missing information cause the most problems? Could you give a recent example?

A: Customers sometimes forget items in cellars or storage rooms, or provide incomplete access information. For example, an item that does not fit in the lift can change the handling time and equipment required.

Q- How do you currently estimate volume in cubic metres, and how do you check that estimate?

A: We estimate volume from the inventory, dimensions and our experience with similar items. We also consider dismantling and how items can be loaded. If the information is uncertain, we request measurements or carry out a further assessment

" We are considering developing a tool that allows customers to upload a short video of each room. The AI then generates an inventory of furniture and boxes, which the customer reviews and corrects. The system then estimates the volume and prepares a preliminary quotation for the moving company to review."

Q- Where could this tool help your current process? Which tasks would still require staff involvement or an on-site visit?

A- It could help us create an initial inventory and reduce the time spent asking customers to list their belongings. Staff would still need to check unusual or fragile items, access conditions and the final quotation. Complex moves might still require an on-site visit

Q- What errors would make the results unusable? What would you need to check before trusting the inventory and volume estimate?

A: Missing large items, counting the same furniture twice or estimating dimensions incorrectly could significantly affect the quotation. Before using the results, we would check that all rooms and storage areas are covered, confirm quantities and review uncertain measurements. The tool should clearly flag items that need verification

Q- What practical concerns could prevent your company or customers from using it?

Would your company consider paying for such a tool? Why or why not? If yes, what pricing arrangement or range would be reasonable?

A- We would consider paying if a trial showed that the tool saves staff time without increasing quotation errors. A payment per completed assessment could be suitable at first. A subscription might make sense for regular use. We would need to compare the price with the time saved and the amount of checking still required

The interview suggests that quotation preparation depends on complete inventories, reliable dimensions and accurate access information. Missing items and uncertain measurements can require further checks. An AI-assisted video tool could help produce an initial inventory, but staff would still need to review uncertain items, access conditions and the final quotation. Willingness to pay was conditional on demonstrated time savings without increased quotation errors. No specific preparation time, cost or acceptable price was provided, so these remain to be quantified.